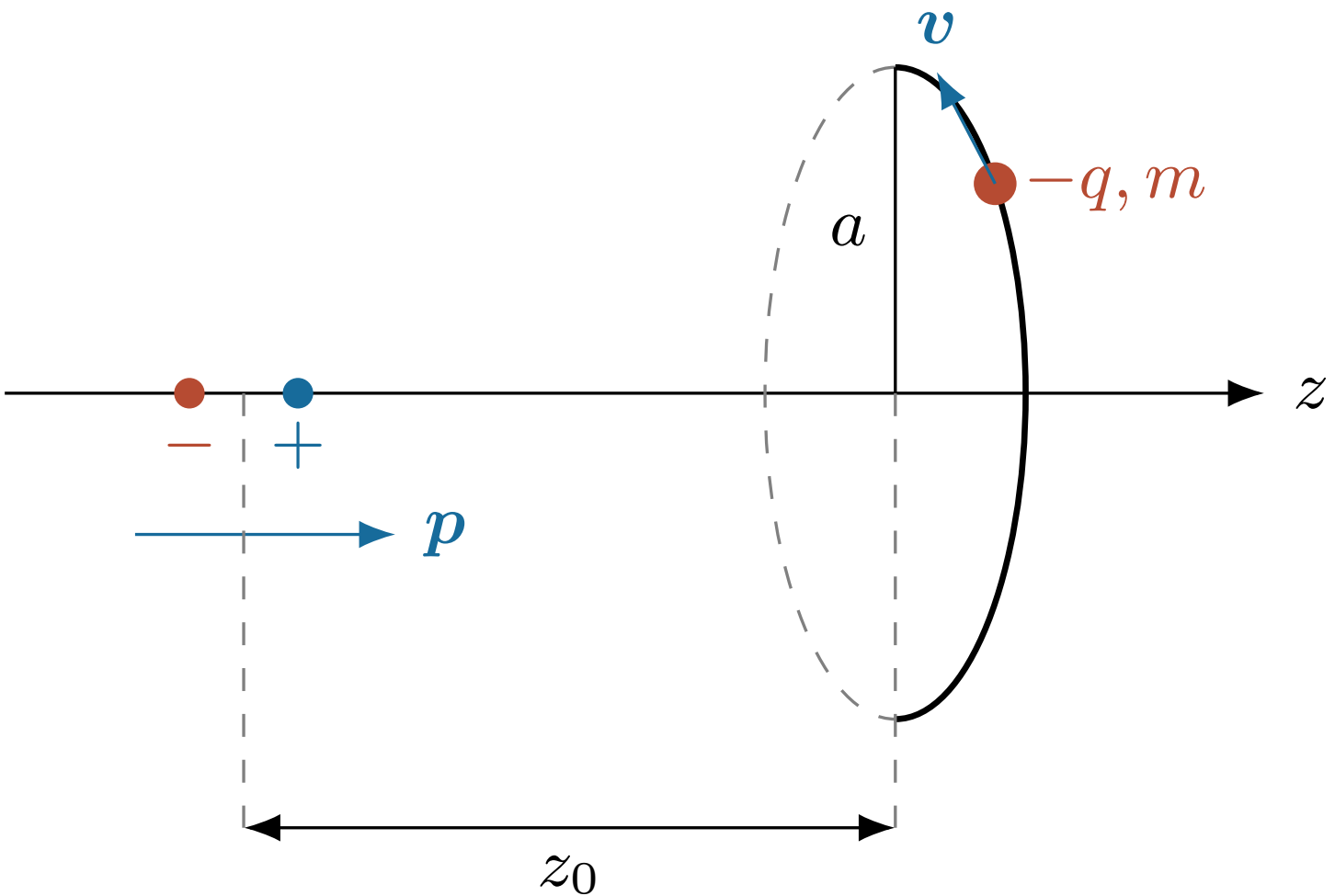
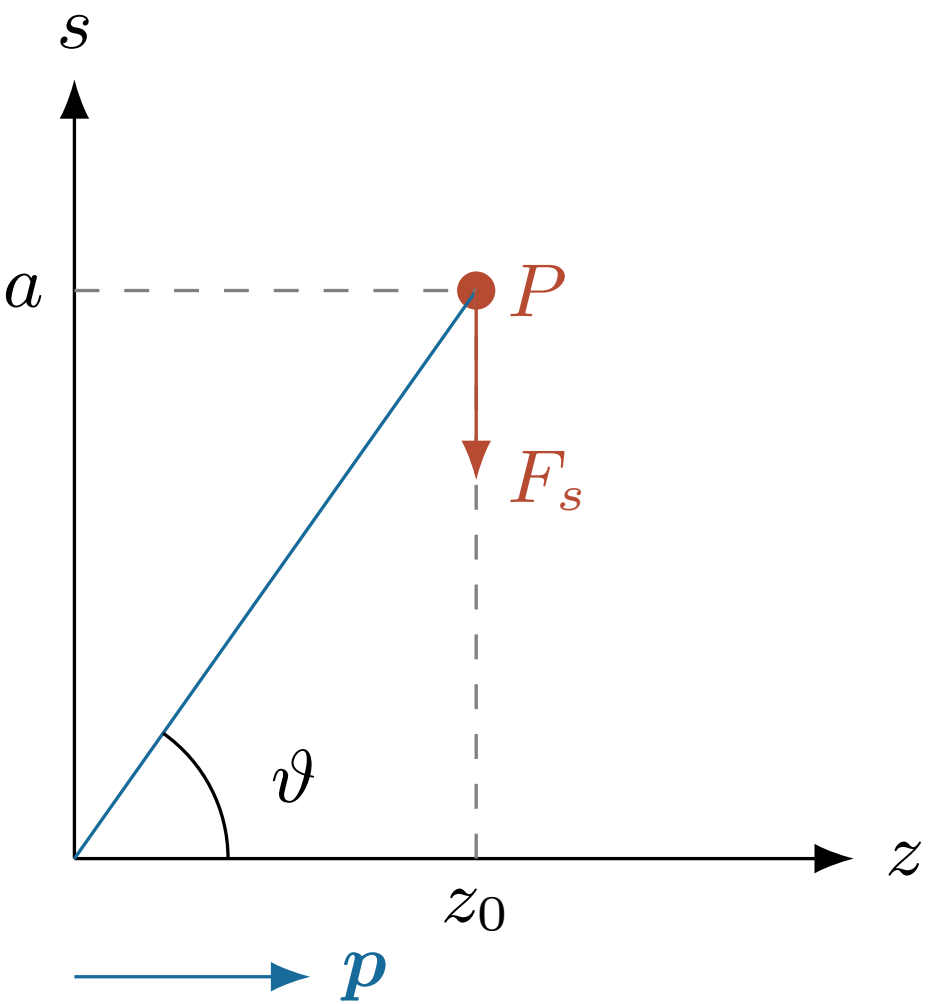
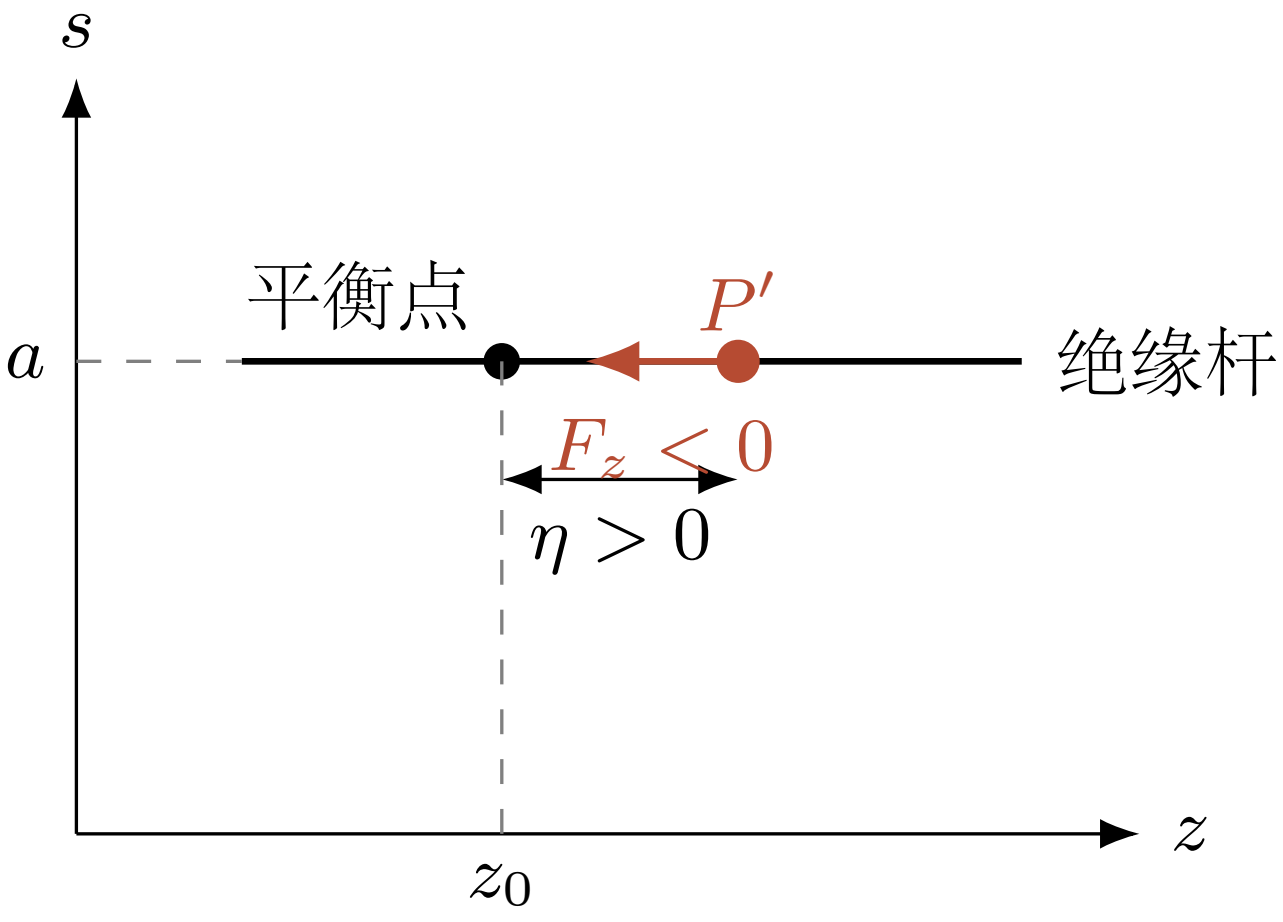
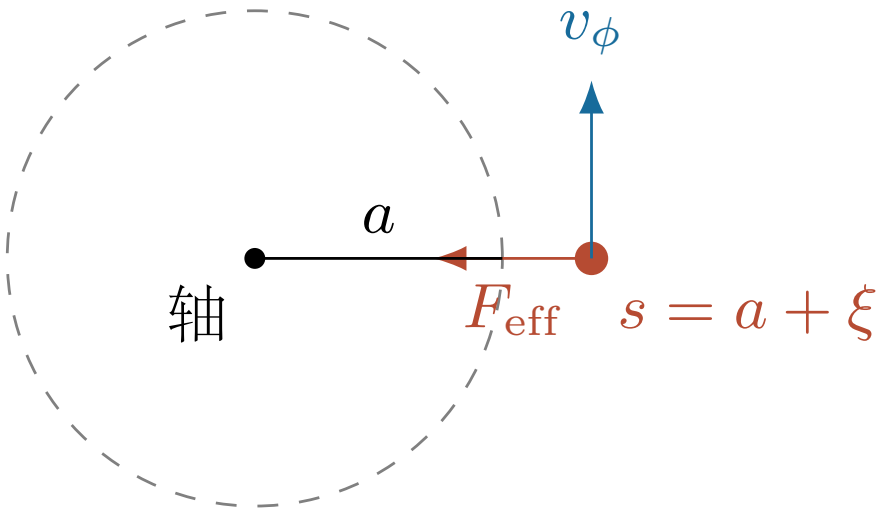
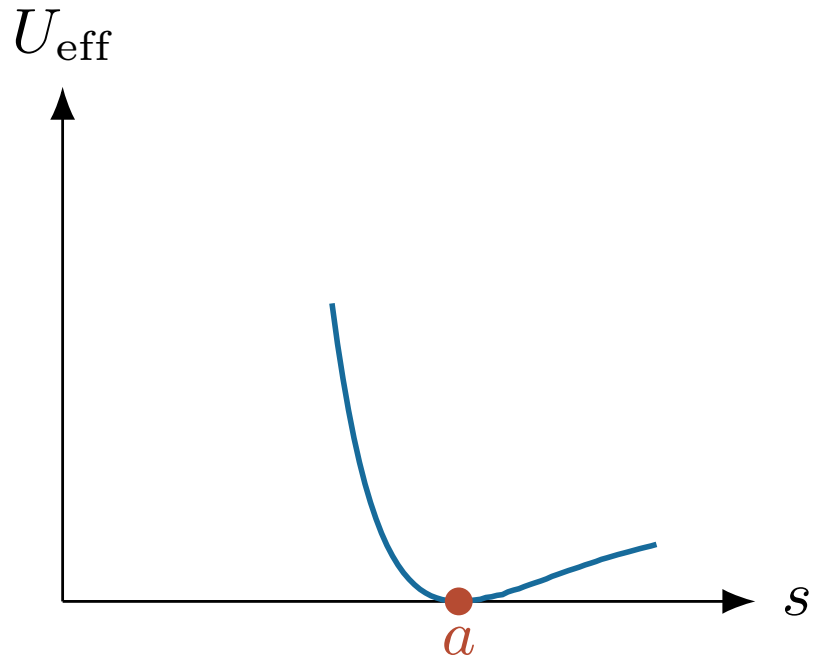

$\boldsymbol{v}$
$-q, m$
a
z
$-$
$+$
$\boldsymbol{p}$
z_0



$$a^2 = 2z_0^2, \quad F_z = 0$$


s
a
平衡点
P′
绝缘杆
F_z < 0
η > 0
z
z_0





固定 z_0，$\ell = msv_\phi$ 守恒

圆轨道是局部最低点